PRACTICAL PROJECT ASSIGNMENT

STUDENT MANAGEMENT SYSTEM

Bootstrap 5 + JSP + Java Servlet + JDBC + MySQL

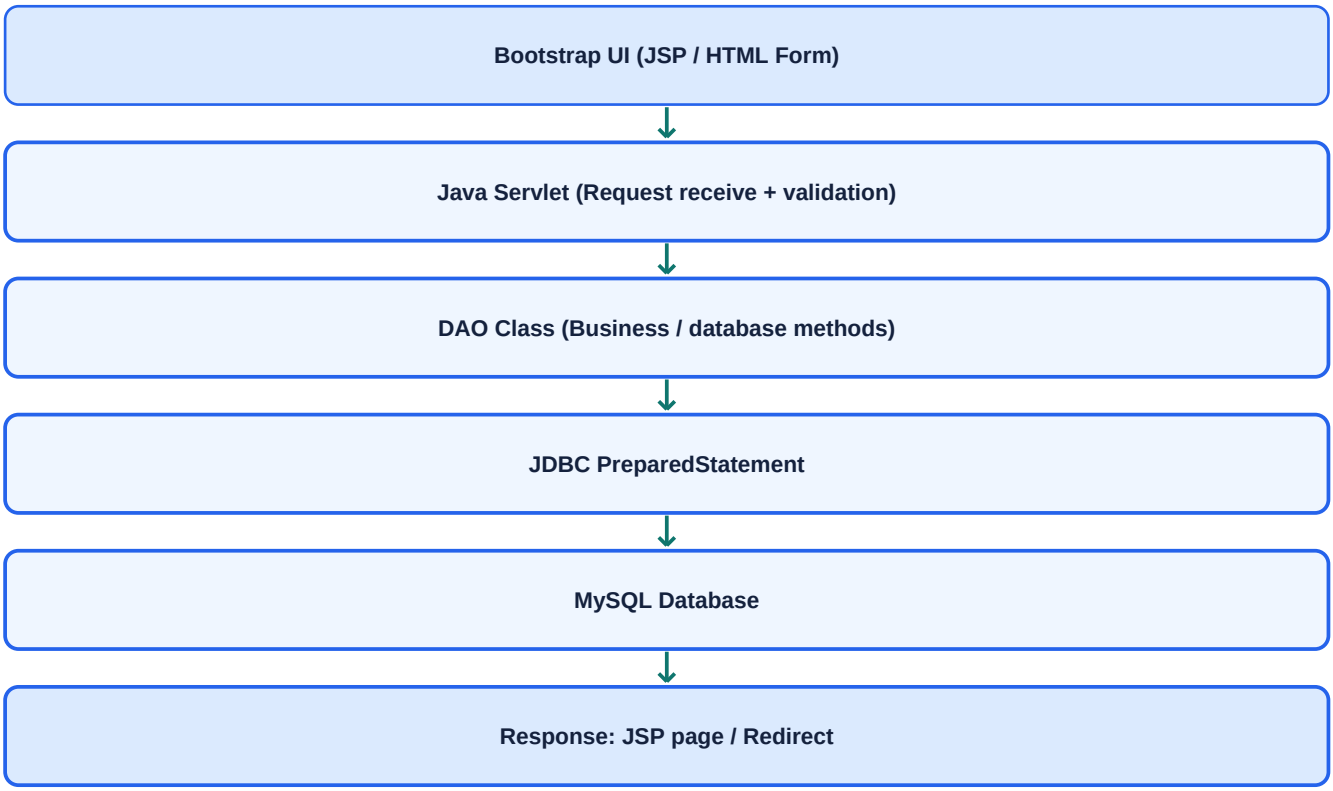
Project Level	Suggested Time	Total Marks
Beginner to Intermediate	5–7 Hours	100 Marks

Is project ka goal

Is assignment mein aap ek chhota Admin Dashboard banayenge. Admin student ko add, list, edit, update, delete aur search kar sakega. Aap seekhenge ki Bootstrap form se data Servlet tak kaise jaata hai, Servlet DAO ko kaise call karta hai aur JDBC ke through MySQL mein data kaise save hota hai.

Important: Sirf code likhna kaafi nahi hai. Har student ko request ka flow, database query aur output apne words mein explain karna hoga.

PROJECT LEARNING FLOW



Student Name: _____ Batch: _____

Submission Date: _____

1. Project Requirements

Build a responsive Student Management System using Bootstrap 5. The application must work on desktop and mobile screens.

Required features

Feature	What should happen?	Servlet / URL
Add Student	New student details database mein save hon.	POST /addStudent
View Students	Database ke saare students table mein dikhein.	GET /students
Edit Student	Selected ID ka existing data form mein aaye.	GET /editStudent?id=ID
Update Student	Changed details same student ID par update hon.	POST /updateStudent
Delete Student	Confirmation ke baad selected record delete ho.	POST /deleteStudent
Search Student	Name, email ya course se record search ho.	GET /searchStudent

2. Technology Stack

Technology	Responsibility
HTML5 + CSS3	Page structure aur custom styling
Bootstrap 5	Responsive navbar, cards, forms, buttons, tables, alerts
JSP	Dynamic data browser par display karna
Java Servlet	HTTP request receive karna aur response control karna
JDBC	Java application ko MySQL se connect karna
MySQL	Student records permanently store karna
JavaScript	Delete confirmation aur basic client-side validation

3. Database Setup

MySQL mein database aur students table banayein. Har student ka unique ID hoga. Email unique rakha gaya hai taaki duplicate email save na ho.

```
CREATE DATABASE student_management;
USE student_management;

CREATE TABLE students (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    mobile VARCHAR(15),
    course VARCHAR(100),
    city VARCHAR(100),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Sample records testing ke liye

```
INSERT INTO students (name, email, mobile, course, city)
VALUES
('Aman Sharma', 'aman@example.com', '9876543210', 'Java', 'Jaipur'),
('Neha Verma', 'neha@example.com', '9876501234', 'Full Stack', 'Ajmer');
```

4. Suggested Project Structure

```
StudentManagement/
|-- src/main/java/
|   |-- com.student.model/
|   |   `-- Student.java
|   |-- com.student.dao/
|   |   `-- StudentDAO.java
|   |-- com.student.util/
|   |   `-- DBConnection.java
|   `-- com.student.controller/
|       |-- AddStudentServlet.java
|       |-- ListStudentServlet.java
|       |-- EditStudentServlet.java
|       |-- UpdateStudentServlet.java
|       |-- DeleteStudentServlet.java
|       `-- SearchStudentServlet.java
|
`-- src/main/webapp/
    |-- index.jsp
    |-- students.jsp
    |-- add-student.jsp
    |-- edit-student.jsp
    `-- WEB-INF/
        `-- web.xml (if using deployment descriptor)
```

Class ka kaam samjhein

Class / Page	Responsibility
Student.java	Student ka data hold karne wala model / POJO.
DBConnection.java	MySQL connection create karna.
StudentDAO.java	Insert, select, update, delete aur search queries.
Servlet classes	Request parameters lena, validation aur navigation.
JSP pages	Bootstrap UI aur dynamic student data display karna.

5. CRUD Request-Response Flow

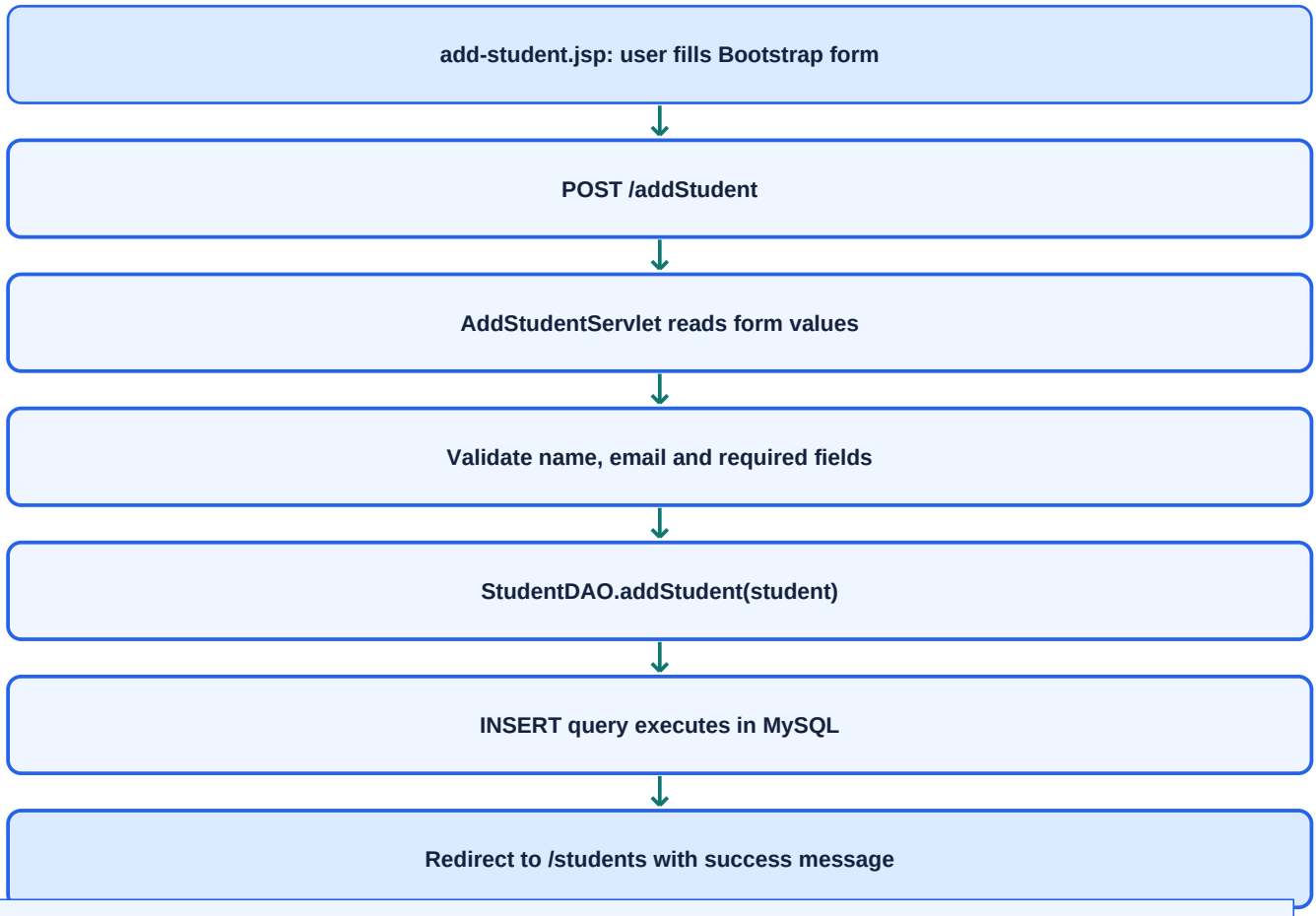
Sabse pehle ye common flow samjhein. Browser mein user action karta hai, Servlet request receive karta hai, DAO database operation karta hai aur phir JSP par result show hota hai.

Forward aur Redirect ka basic difference

forward: Server ke andar request ko JSP tak bhejta hai. Request attributes JSP mein available rehte hain. Example: edit form open karna.

sendRedirect: Browser ko nayi URL request karne ko kehta hai. Example: successful add/update/delete ke baad student list par bhejna.

6. ADD Student Flow

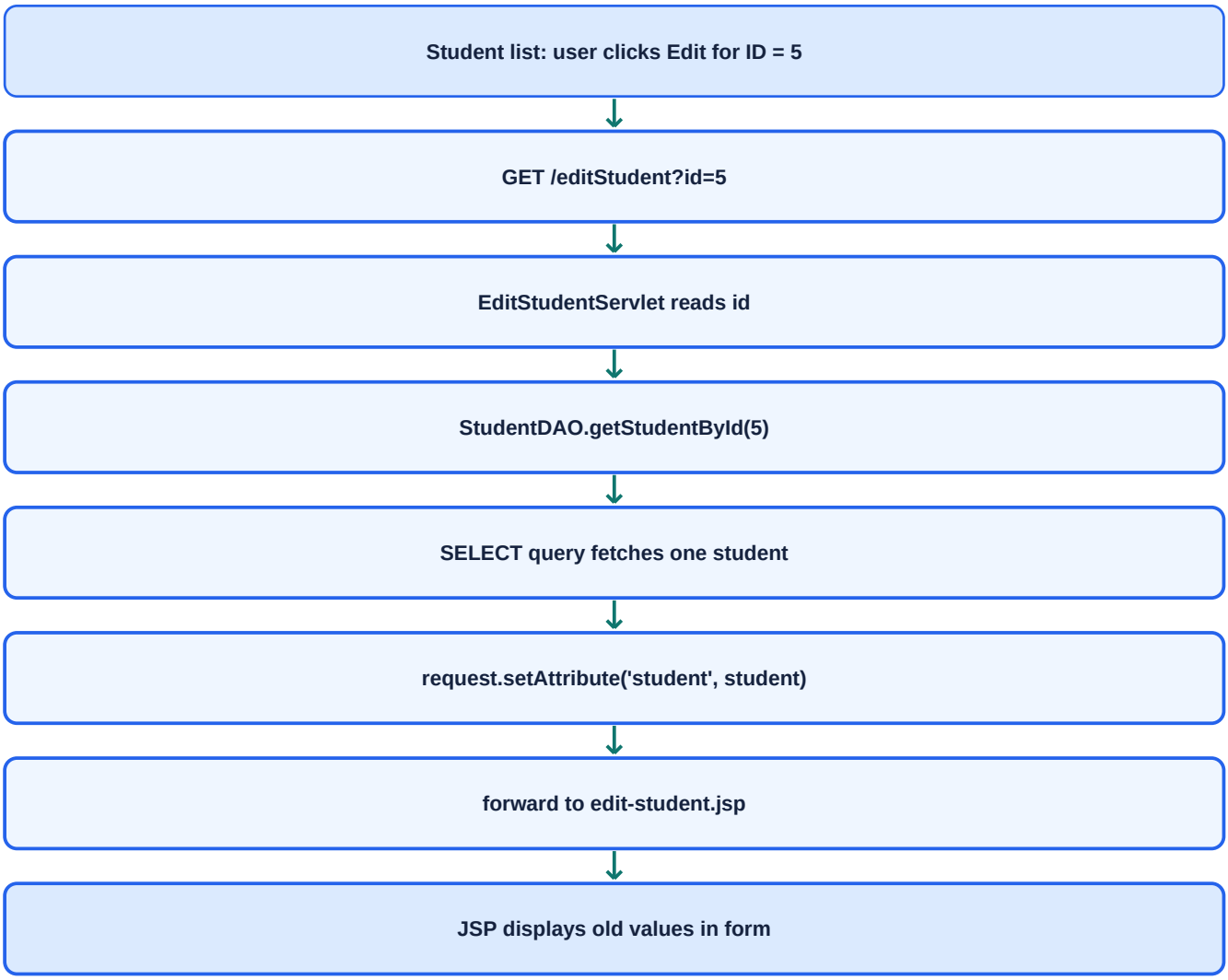


Student ko samajhna hai: Form ka name attribute Servlet ke request.getParameter("name") se match karta hai. DAO database ka kaam karta hai; Servlet mein SQL query directly nahi likhni.

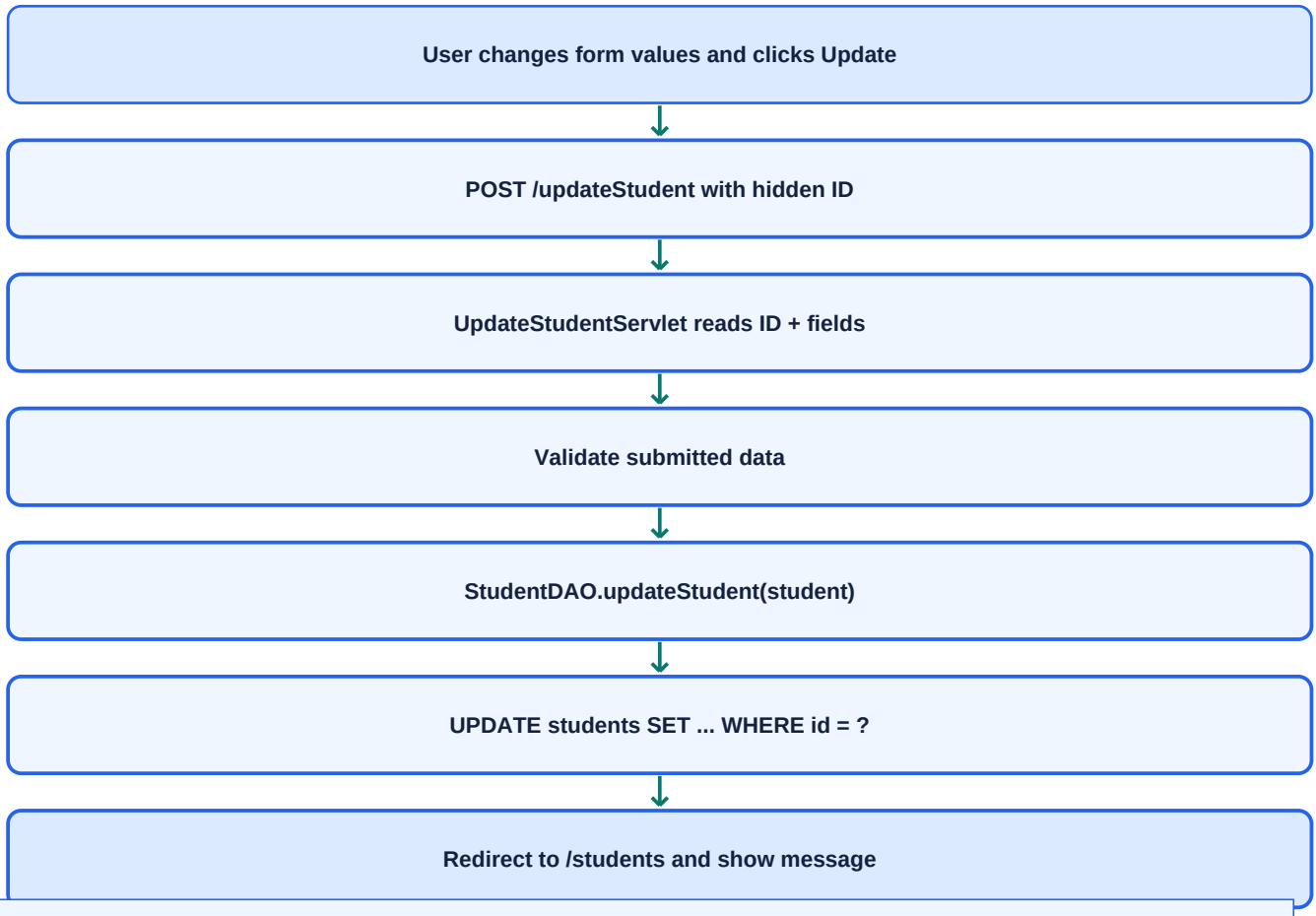
7. EDIT aur UPDATE – Sabse Important Flow

Edit aur Update do alag actions hain. Edit ka kaam record ko fetch karke form mein dikhana hai. Update ka kaam changed values ko database mein save karna hai.

A. EDIT FLOW (GET)



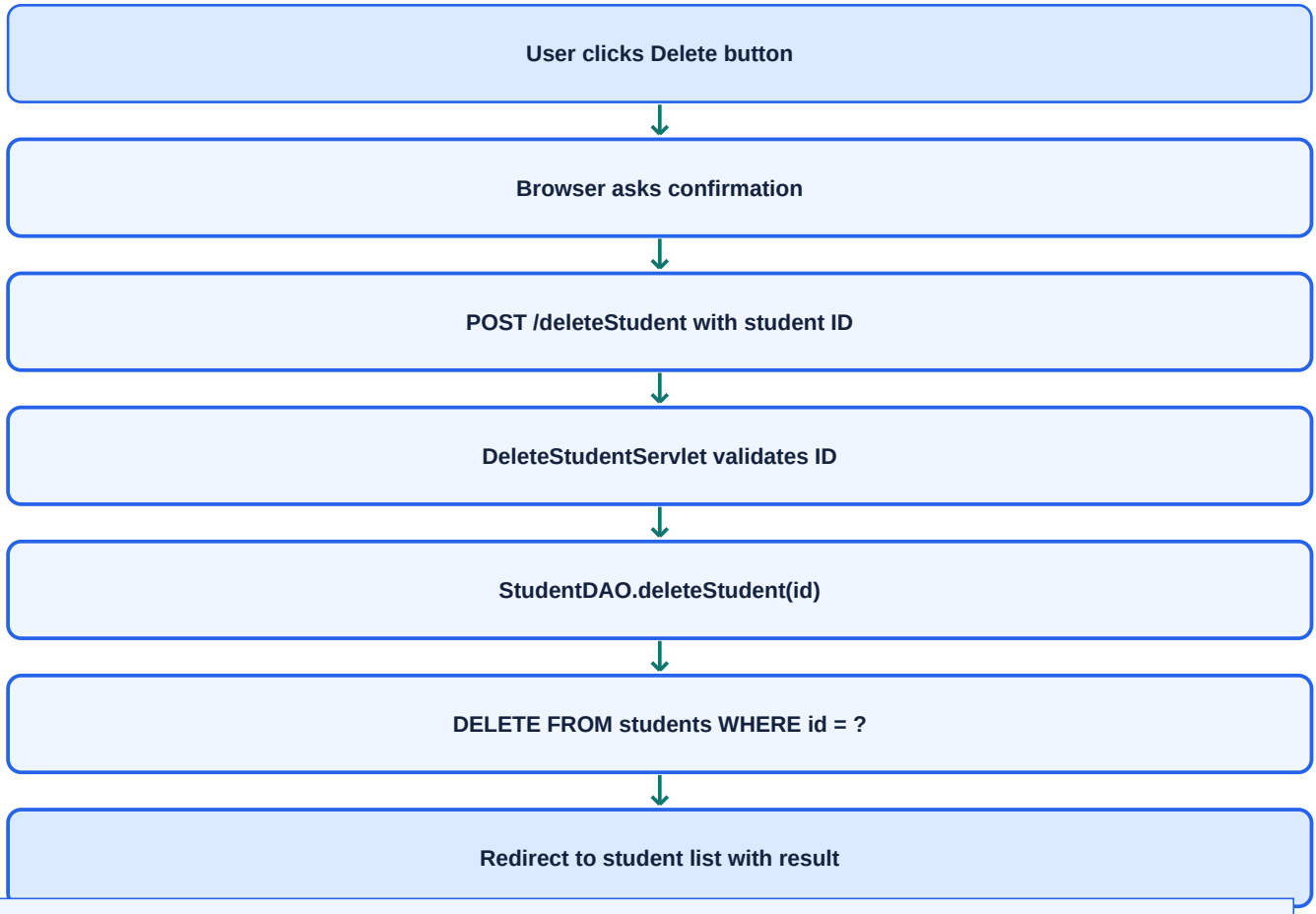
B. UPDATE FLOW (POST)



Important logic: Edit page ke form mein student ID hidden field mein bhejein. Update query mein WHERE id = ? zaroor hona chahiye. WHERE ke bina table ke multiple/all records update ho sakte hain.

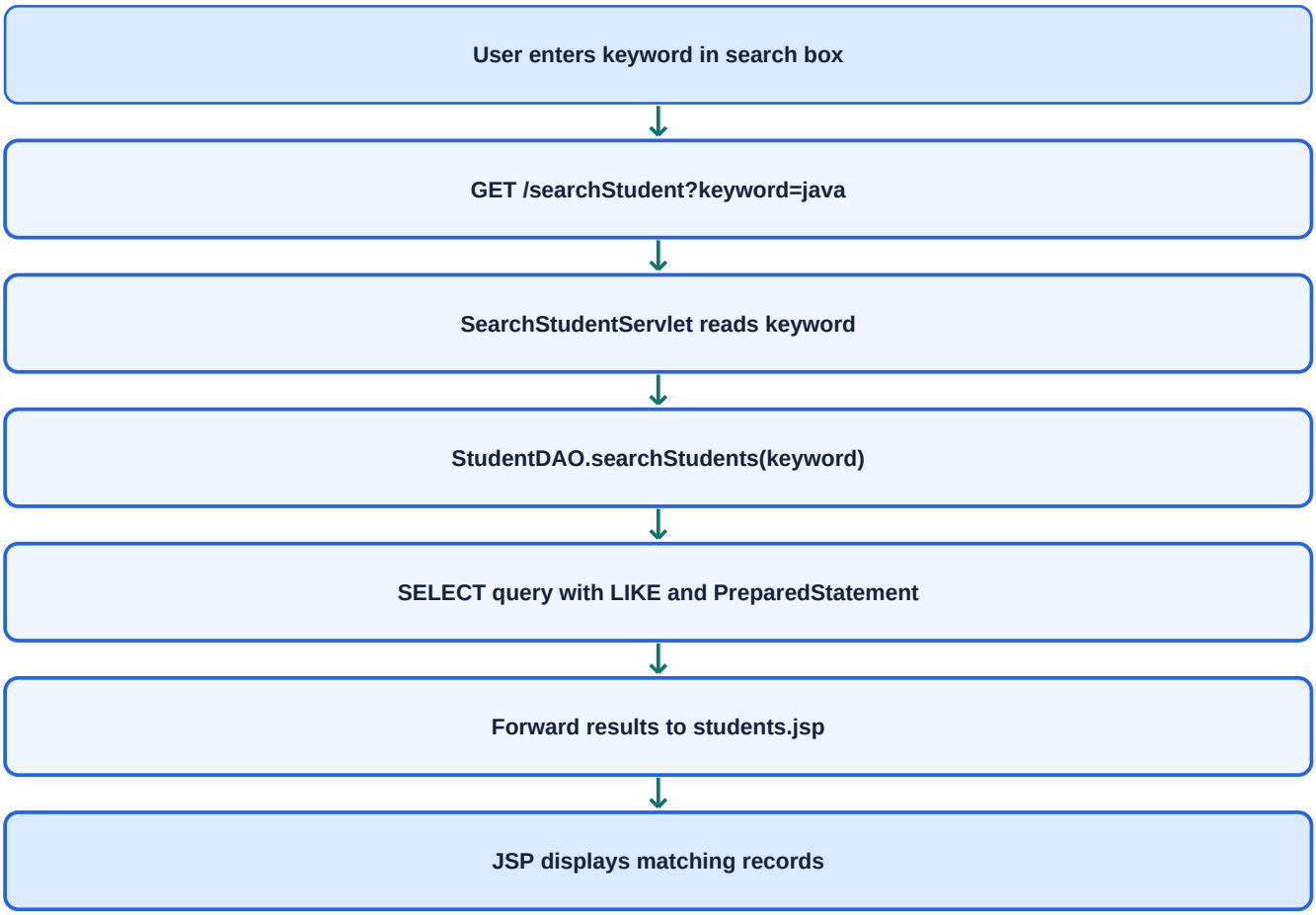
```
<!-- edit-student.jsp: important form idea -->
<form action="updateStudent" method="post">
    <input type="hidden" name="id" value="${student.id}">

    <input type="text" name="name"
           value="${student.name}" required>

    <input type="email" name="email"
           value="${student.email}" required>

    <button type="submit">Update Student</button>
</form>
```

8. DELETE Student Flow



Delete ke liye GET use na karein. Delete ek data-changing action hai, isliye POST use karein. Confirmation helpful hai, lekin server-side validation phir bhi zaroori hai.

9. SEARCH Student Flow



10. Bootstrap UI Requirements

UI ko simple, clean aur responsive rakhein. Bootstrap 5 components use karein:

Page	Bootstrap components
Dashboard	Navbar, sidebar, cards, badges
Student List	table, table-responsive, form-control, btn, badge, alert
Add Student	card, row, col-md-6, form-label, form-control, form-select
Edit Student	card, pre-filled form controls, btn-primary
Delete	btn-danger + JavaScript confirm()

11. DAO Methods – Students Must Implement

```
public interface StudentDAO {
    boolean addStudent(Student student);

    List<Student> getAllStudents();

    Student getStudentById(int id);

    boolean updateStudent(Student student);

    boolean deleteStudent(int id);

    List<Student> searchStudents(String keyword);
}
```

DAO method ka matlab: Servlet method call karta hai, DAO actual SQL/JDBC operation karta hai aur result Servlet ko return karta hai.

12. SQL Logic – Understand Before Coding

```
-- ADD
INSERT INTO students (name, email, mobile, course, city)
VALUES (?, ?, ?, ?, ?);

-- VIEW ALL
SELECT * FROM students ORDER BY id DESC;

-- EDIT: fetch one record
SELECT * FROM students WHERE id = ?;

-- UPDATE: same ID ka record update
UPDATE students
SET name = ?, email = ?, mobile = ?, course = ?, city = ?
WHERE id = ?;

-- DELETE: selected ID only
DELETE FROM students WHERE id = ?;

-- SEARCH
SELECT * FROM students
WHERE name LIKE ? OR email LIKE ? OR course LIKE ?;
```

13. Bootstrap 5 CDN

```
<!-- Put inside the <head> of each JSP page -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
      rel="stylesheet">

<!-- Put before closing </body> -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.min.js"></script>
```

Internet access na ho to Bootstrap files locally download karke project mein include kar sakte hain. JSP mein JSTL use kar rahe hain to JSTL dependency aur taglib declaration bhi configure karein.

14. Important Development Rules

- JDBC queries ke liye PreparedStatement use karein.
- Connection, PreparedStatement aur ResultSet ko try-with-resources se close karein.
- Servlet mein input validate karein; sirf HTML required attribute par depend na karein.
- Database password source code mein public repository par commit na karein.
- Add, Update, Delete ke baad redirect use karein (PRG pattern).
- Error ko silently ignore na karein; useful message log aur display karein.

15. Testing Checklist

Har feature ko normal aur invalid input dono se test karein.

Test	Expected result	Done
Add valid student	Record database mein insert ho.	■
Add empty name	Validation error aaye.	■
Add duplicate email	Duplicate email reject ho.	■
View students	All records table mein dikhein.	■
Edit existing ID	Correct student ka old data form mein aaye.	■
Update student	Sirf selected ID ka record change ho.	■
Edit invalid ID	Not found / 404 type message aaye.	■
Delete student	Sirf selected record delete ho.	■
Search by name/course	Matching records show hon.	■
Mobile screen	Layout horizontal overflow ke bina usable ho.	■

16. Viva Questions

1. Edit aur Update mein kya difference hai?

2. doGet() aur doPost() kab use karte hain?

3. request.getParameter() kya return karta hai?

4. sendRedirect() aur RequestDispatcher.forward() mein kya difference hai?

5. PreparedStatement kyun use karte hain?

6. executeQuery() aur executeUpdate() ka use kya hai?

7. Update query mein WHERE id = ? kyun zaroori hai?

8. DAO pattern ka benefit kya hai?

9. JSP mein request attribute kaise read hota hai?

10. Post-Redirect-Get (PRG) pattern kya hai?

17. Marks Distribution

Area	Marks
Database design + JDBC connection	10
Bootstrap responsive UI	15
Add + View functionality	15
Edit + Update functionality	20
Delete + Search functionality	15
Validation + error handling	10
Code structure + DAO separation	5

Flow explanation + viva	10
TOTAL	100

Final submission: Complete source code, SQL file, screenshots of all CRUD operations, and a 3–5 minute explanation of the request-response flow. Student ko project bina code dekhe explain karna aana chahiye.